Portfolio Risk Assessment

As the Chief Risk Officer of ***InsurAIR***, an insurance company operating in the town of AIRton, your role is to perform portfolio risk assessment for the company's insurance portfolio with respect to flooding which happens to the governing natural hazard in the area. When it comes to risk assessment, the framework involves multiple components and steps and can be described using the diagram below:

Figure 1: Components of a catastrophe model

The first step involves creating many stochastic flood events and the local intensities (in this case, flood depths) that each of these events produce at a given location. This can be thought of analogous to assembling a ground motion suite and the local intensities (peak ground acceleration, peak ground velocity etc.) that each of the ground motion produces spatially. The creation of the events and the associated local intensities together form the hazard module.

Exposure data typically includes information regarding the individual risks/assets/buildings that are a part of the insurance portfolio. Information includes location (street address or other geographic information such as latitude, longitude), occupancy of the building, construction type, number of stories, year of construction, valuation of the building (typically called replacement value) for the physical assets and also per diem loss of revenue if the business were inoperable. Please refer to https://unicede.air-worldwide.com/ for a complete description of all information that is a part of an exposure portfolio.

The engineering/damage estimation component then calculates the impact of a given event and the intensity it produced at the location of an asset in terms of loss i.e. damage or repair cost. Damage or vulnerability functions translate the intensity of hazard affecting a structure to a damage ratio which is the ratio of damage or repair cost to the replacement value (cost to rebuild the entire structure). The following article is a good resource for you to read more on the topic:

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https://www.air-worldwide.com/publications/air-currents/2017/modeling-fundamentals-anatomy-of-a-damage-function/

The final step in the process is loss estimation. Individual location level losses for building damage, loss of business for a given event can then be obtained by multiplying the respective damage ratios to their corresponding replacement values. Once an event loss table is created, portfolio risk assessment can be performed by using various risk metrics such as average annual loss (AAL), specific return period losses alternately referred to as losses with a certain probability of exceedance, value at risk (VaR), to mention a few. The following articles shed some light on these metrics and how they can be obtained once we calculate individual event level losses at all locations within an exposure portfolio:

https://www.air-worldwide.com/publications/air-currents/2013/Modeling-Fundamentals--What-Is-AAL-/

https://www.air-worldwide.com/publications/air-currents/2014/Modeling-Fundamentals--Evaluating-Risk-Measures/

Your task is to write a computer program in a programming language of your choice to help perform portfolio flood risk assessment. The subsequent sections describe the inputs and the desired outputs.

Description of Inputs

InsurAIR Exposure Portfolio

Information regarding every risk that InsurAIR insures is in the file titled ***InsurAIR_ExposurePortfolio_2021.csv***.

The town of AIRton is a well-planned town and measures 9 km x 9 km in its spatial footprint. The town is therefore idealized in 1 km grids as shown in Figure 2 below. This layout can be characterized using a cartesian coordinate system with the origin positioned at the lower left. Distances measured along the horizontal axis in km can be perceived as the longitude and distances measured along the vertical axis in km can be perceived as the latitude.

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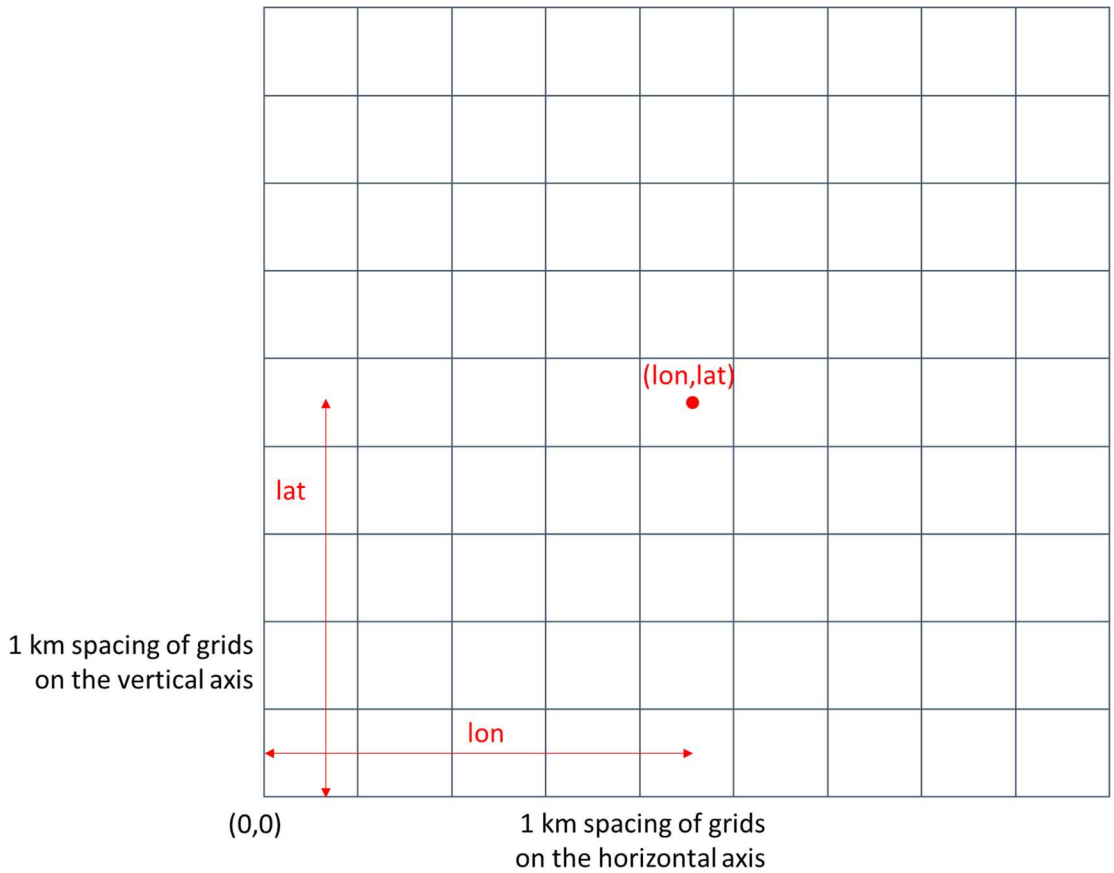

Figure 2: Layout of the town of AIRton

Here is a description of the columns in the aforementioned file.

Location ID – unique identifier for a given location

Location Lat – this can be described as the y-coordinate in the cartesian coordinate system where the origin is at the bottom left. More details in the next section where intensity files are described

Location Lon – this can be described as the x-coordinate in the cartesian coordinate system where the origin is at the bottom left. More details in the next section where intensity files are described

Occupancy – describes the utility of the building. 302 denotes a single-family home while 315 denotes a building used for professional, technical and business services

Construction – describes the material used in constructing the building. 101 represents wood frame construction while 131 denotes reinforced concrete construction

Number of Stories – denotes the height of the building. 991 refers to low rise buildings less than 3 stories tall, 992 represents mid-rise buildings between 4 and 7 stories and

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993 represents high-rise buildings 8 stories and beyond

Building Replacement Value - cost to rebuild the entire physical building in U.S. dollars

Perdiem Loss of Business Revenue/Perdiem Additional Living Expense – This is the daily loss of revenue for commercial buildings when the buildings cannot be used due to damage. In the case of residential buildings, this is the cost the InsurAIR incurs to provide alternate accommodation if the building is deemed to be not livable.

Flood Intensity File

The stochastic event set has 100 flood events. The flood intensity file shows the location level flood depth in feet for each of the 100 flood events at the corners of the grid cells shown in Figure 2 above. The event level flood depth footprints are in the file titled ***FloodIntensityFile_AIRton_2021.dat***.

Here is a description of the columns in the file

Year: year of the event

EventID: event id identifier

GridID: grid identifier ranging from 0000 to 0909. Where the first 2 digits indicate the XX value and the last digits indicate the YY of the coordinate system

Intensity: flood depth in feet

As you will notice, the exposure points i.e., building locations can fall within a 1 km x 1 km grid cell and the intensity values provided are at the grid corners. You can consider the average intensity of the four corners of the grid cells as the intensity at the building location.

Damage Function Files

Separate damage functions exist to characterize physical damage resulting from flood to the building (typically referred to as building or coverage A damage) and loss of revenue/additional living expense (typically referred to as business interruption/time element or coverage D damage).

Building or coverage A damage function relates the damage ratio (ratio of repair cost to replacement value) to the flood depth. Time element or coverage D damage functions relate number of days of downtime to the building damage ratio. In essence, one needs to determine physical damage to the building (coverage A damage function) to then calculate the associated downtime (in number of days) to the same building. Figures 3 and 4 show representative building and time element damage functions.

The building and time element damage functions that are specific to the risks insured in InsurAIR's portfolio are provided in the files titled

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InsurAIR_BuildingDamageFunction.xlsx and
InsurAIR_TimeElementDamageFunction.xlsx

Each of these files have 2 worksheets in them: ***Mapping*** and ***DamageFunction***.

The mapping sheet indicates which damage function to use depending on the combination of occupancy construction and number of stories. The Damage Function sheet provides the actual damage function for all IDs listed in the Mapping sheet.

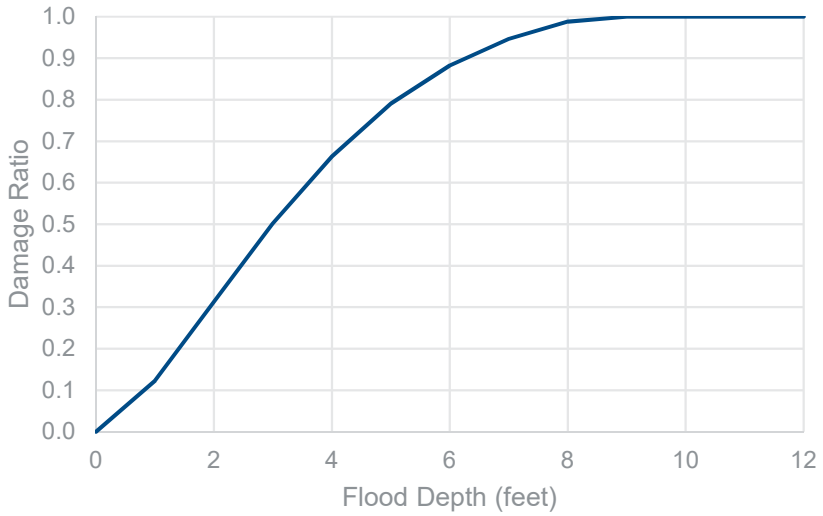

Figure 3: Sample building damage function relating building damage ratio to the flood depth

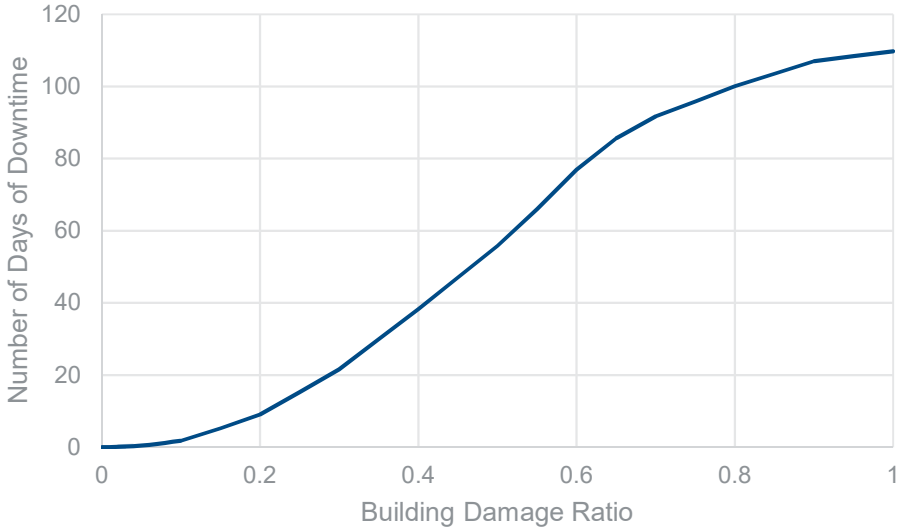

Figure 4: Sample time element damage function relating time element damage measures as number of days of downtime to the building damage ratio

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Please note the following as you perform your calculations:

Damage functions are characterized for specific values of flood depth (in the case of building damage functions) and building damage ratio (in the case of time element damage functions). For intermediate values of input, consider interpolating between the nearest values to obtain the damage ratios.

Desired Outputs

1. A well commented program that reads in the aforementioned input files, performs the calculations described in this section and writes out the desired output files. If any assumptions are made to perform calculations, please document the same in the body of the program. The only deliverable is the program itself and output files described below
2. The program should read in the exposure, intensity and damage function files. Once these files are read in, the program should then calculate the location level losses for building damage (coverage A) and time element damage (coverage D) for every event in the intensity file. In essence it has to loop through all the events and location combinations to create and output an event and location loss table. The table whose format is shown in the Table 1 below should be written out to a file titled:
   LocationEventLossTable_InsurAIR_LastNameFirstName.dat

Table 1: Event Location Loss Table Desired Format

Event Number	Location 1 – Building Loss	Location 1 – Time Element Loss	Location 1 – Total Loss (Building + Time Element Loss)	…	…	InsurAIR portfolio – Building Loss	InsurAIR portfolio – Time Element Loss	InsurAIR portfolio – Total Loss (Building + Time Element Loss)
1								
2								
…								
100								

Please note that the portfolio level loss is the sum of losses to all affected locations within the insurance portfolio.

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3. The program should create a portfolio level aggregate exceedance probability curve (after adding all event level losses in a given year) and write out the same in a file titled: ***EPCurve_InsurAIR_LastNameFirstName.dat.*** For details as to how to create an exceedance probability curve, please refer to the following article: https://www.air-worldwide.com/publications/air-currents/2013/Modeling-Fundamentals--What-Is-AAL-/

   Table 2 shows the format of this deliverable:

Table 2: Exceedance probability curve format

Exceedance probability	Return period (1/exceedance probability)	InsurAIR portfolio – Total Loss (Building + Time Element Loss)
AAL	AAL	
0.5	2	
0.2	5	
0.1	10	
0.04	25	
0.02	50	
0.01	100	

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